TAKE-NO-CHANCES APPROACH

Big changes between Chandrayaan-2 and 3

NAVIGATION

Multiple sensors and instruments to check altitude, velocity and descent to log accurate, real-time data on the descent path. These included:

- **Altimeters**: To measure altitude variations, upgraded laser and radio frequency-based altimeters used
- **Velocimeters**: To gauge velocity, there was an all-new laser Doppler velocimeter apart from an improved lander horizontal velocity camera
- **Inertial system**: For flight path, there was improved laser gyro-based inertial referencing and accelerometer package
- Improved lander hazard detection and avoidance camera to ensure landing at suitable spot

PROPULSION SYSTEM

- Four instead of five 800N throttleable liquid engines
- Slosh mitigating by anti-slosh baffle in the propellant tanks
- Rigorous integrated testing of engines, valves, and electronics for better performance
- 182 kg more fuel carried to give greater manoeuvrability

GUIDANCE CONTROL SYSTEMS

- Improved software designed for powered descent trajectory
- Multiple mission paths built for soft-landing for contingencies
- Thrust and angle continuity throughout the descent by appropriate selection of guidance algorithm
- Instantaneous thrust corrections: Done for deviations unlike in Chandrayaan-2

LUNAR EXPERIMENTS

BY LANDER

- Radio Anatomy of Moon Bound Hypersensitive Ionosphere and Atmosphere: To measure the near surface plasma (ions and electrons) density and its changes with time
- Chandra's Surface Thermophysical Experiment: To measure the thermal properties of the lunar surface
- Instrument for Lunar Seismic Activity: To measure seismicity around the landing site and delineating the structure of the lunar crust and mantle
- Laser Retroreflector Array: To understand the dynamics of the moon system

BY ROVER

- Laser-induced Breakdown Spectroscope: To derive the chemical composition and infer mineralogical composition of lunar surface
- Alpha Particle X-ray Spectrometer: To determine the elemental composition, including magnesium, aluminium, silicon, ferrous and other minerals of lunar soil and rocks, around the landing site